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**Inoue**

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(54) **SERVER DEVICE, MONITORING METHOD,  
AND NON- TRANSITORY  
COMPUTER-READABLE MEDIUM**

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(57) **ABSTRACT**

An object is to provide a server device capable of performing analysis on a terminal device applying redundancy to a communication path and simultaneously communicating with a plurality of access points. A server device (10) according to the present disclosure includes: a communication unit (11) receiving, from a sensor device that collects packets transferred between a wireless terminal and a first access point/a second access point, sensor information generated based on collected packets; a management unit (12) managing the first access point and the second access point in association with the wireless terminal when the wireless terminal communicates with the first access point and the second access point by applying redundancy to the communication path; and an analysis unit (13) generating information on a communication state of redundant communication paths by using sensor information regarding the first access point and the second access point.

**9 Claims, 7 Drawing Sheets**

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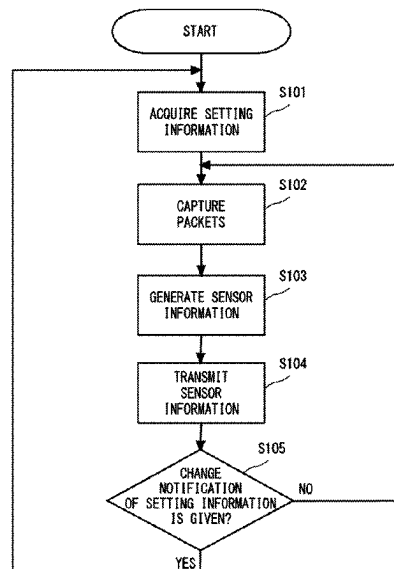
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**H04W 40/12** (2009.01)

(52) **U.S. Cl.**  
CPC ..... **H04W 40/12** (2013.01)

(58) **Field of Classification Search**  
None

See application file for complete search history.



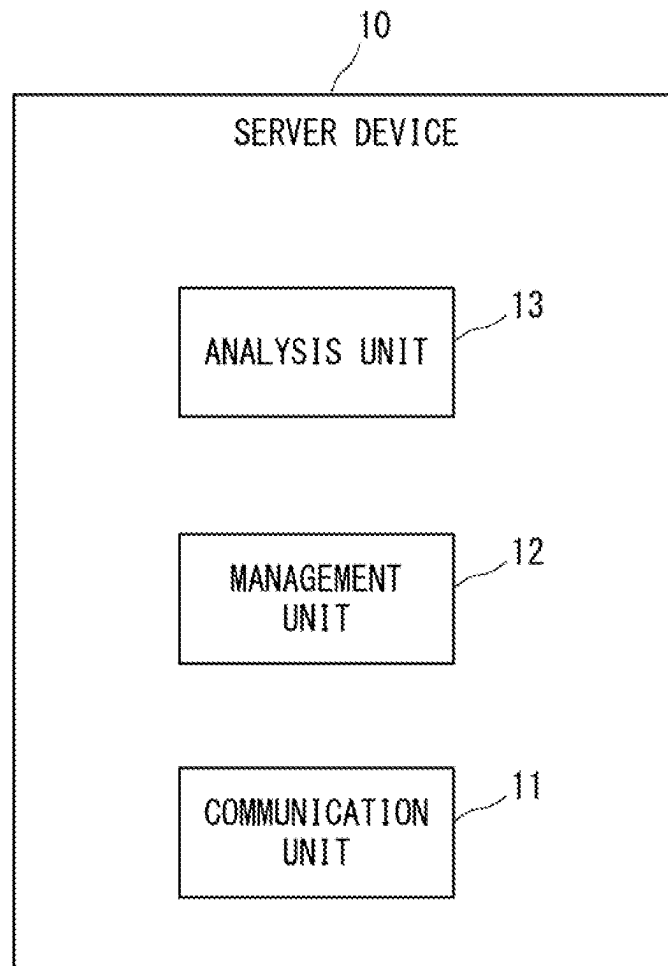


Fig. 1

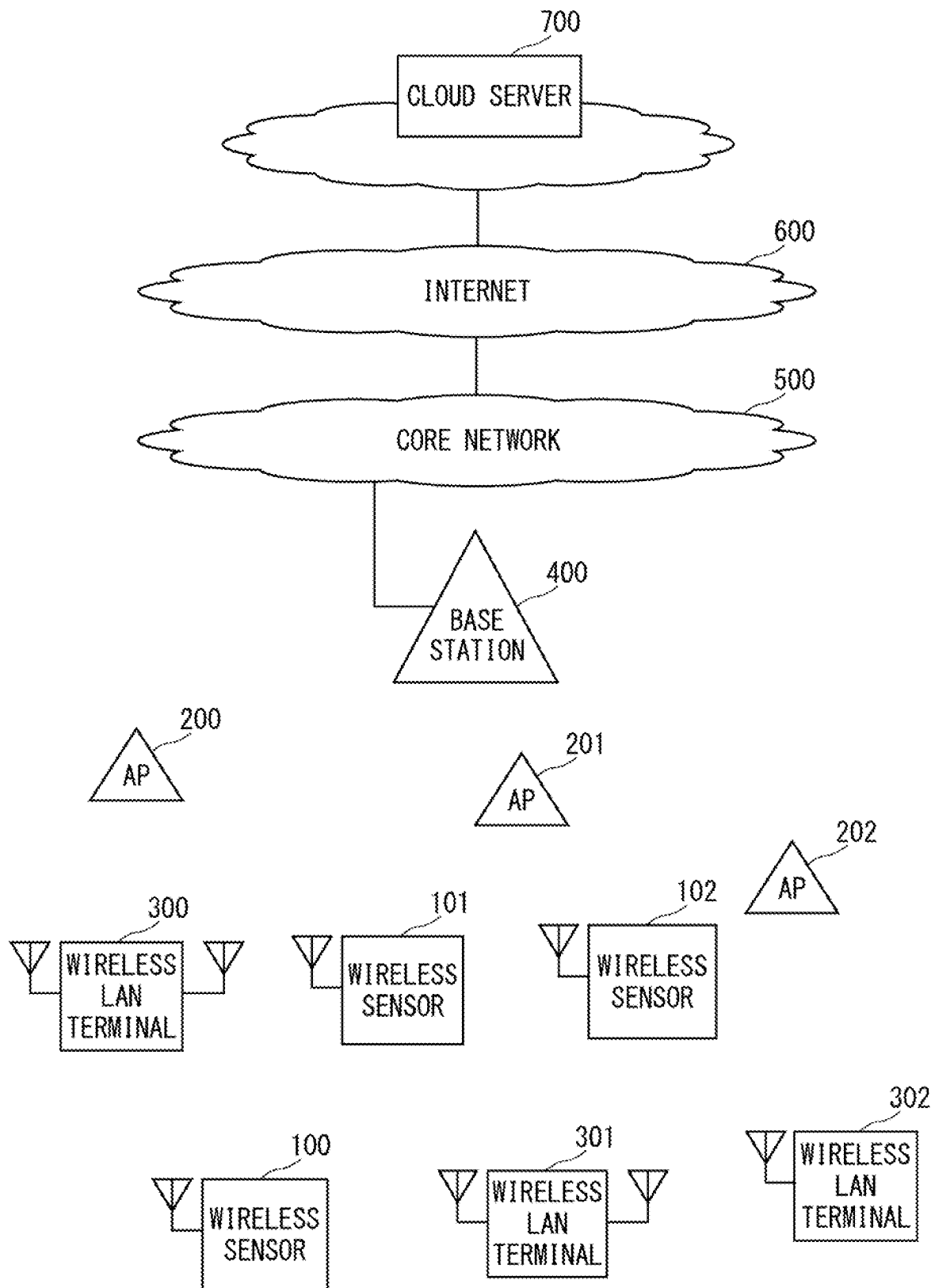


Fig. 2

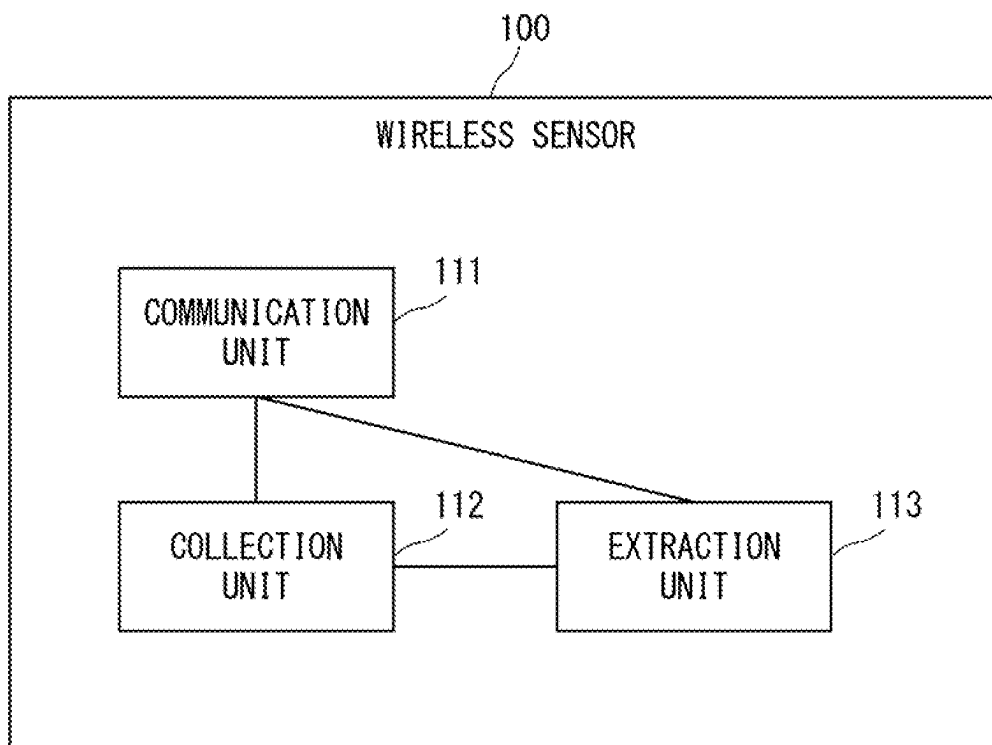


Fig. 3

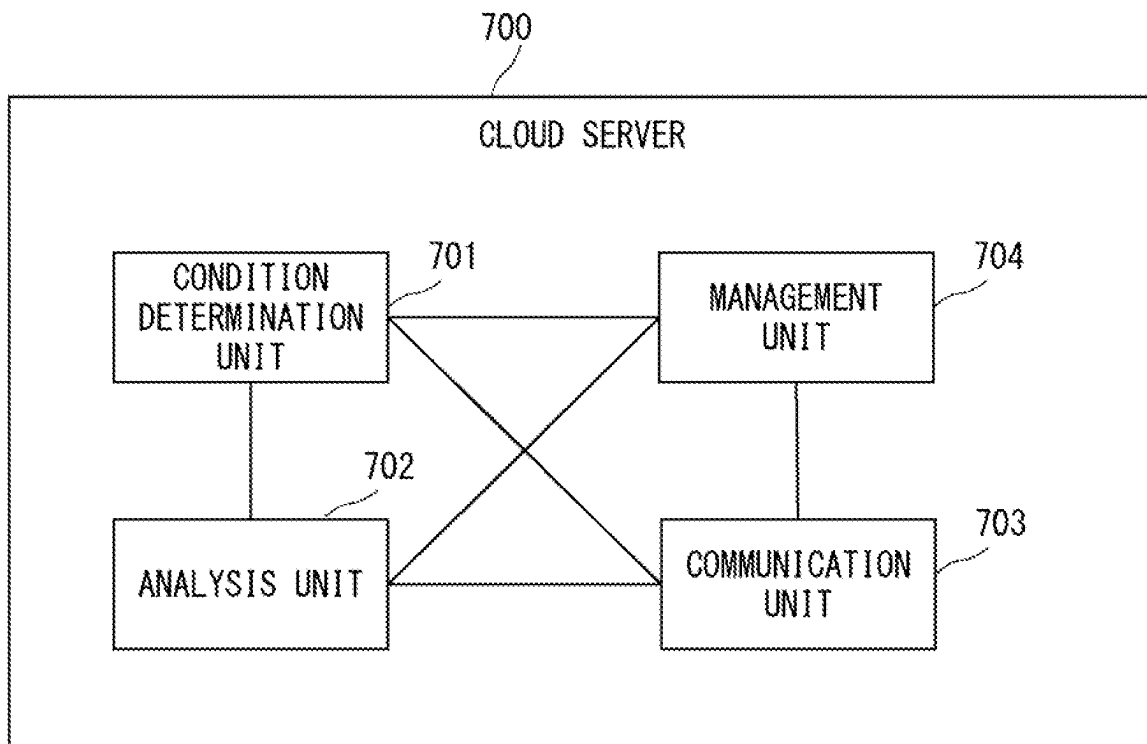


Fig. 4

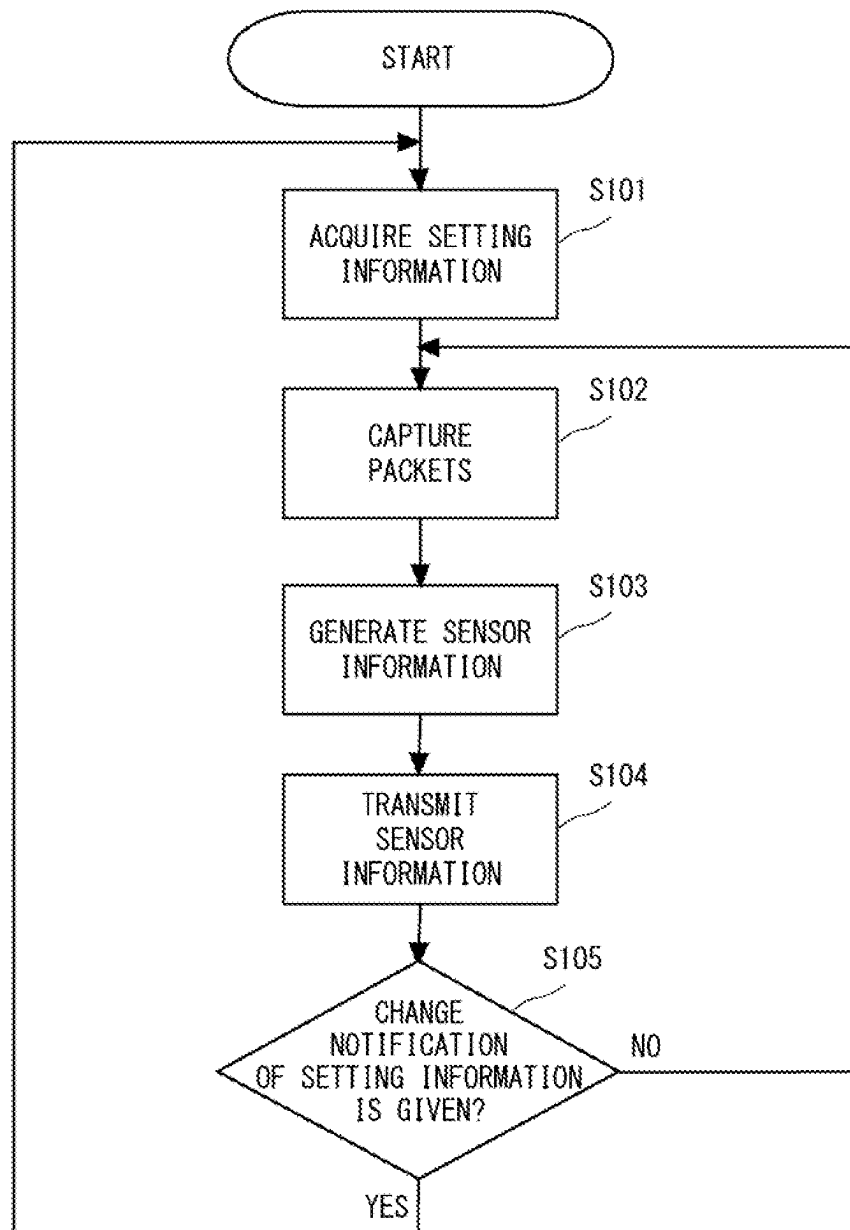


Fig. 5

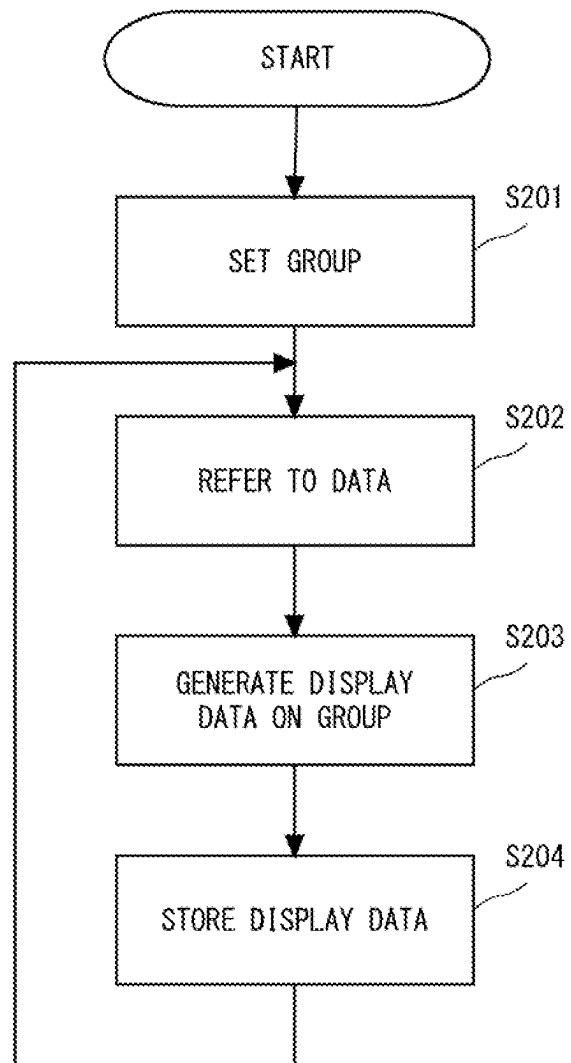


Fig. 6

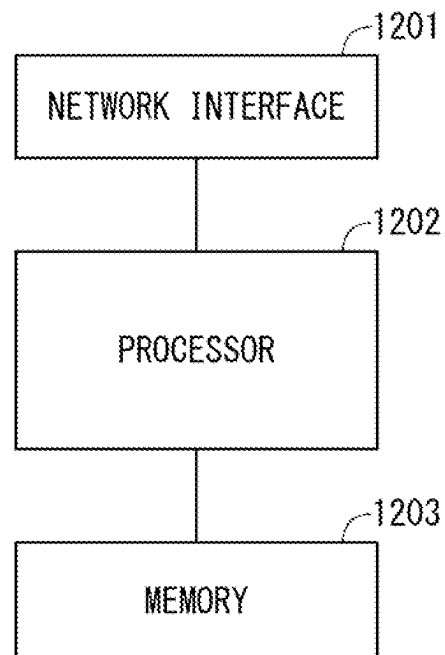
10, 100, 700

Fig. 7



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# SERVER DEVICE, MONITORING METHOD, AND NON- TRANSITORY COMPUTER-READABLE MEDIUM

This application is a National Stage Entry of PCT/JP2021/024878 filed on Jun. 30, 2021, which claims priority from Japanese Patent Application 2020-114079 filed on Jul. 1, 2020, the contents of all of which are incorporated herein by reference, in their entirety.

## TECHNICAL FIELD

The present disclosure relates to a server device, a monitoring method, and a program.

## BACKGROUND ART

In recent years, a cloud server maintains or improves quality of a wireless communication system by managing communication quality in the wireless communication system. The cloud server acquires information regarding the communication quality from a wireless communication terminal in the wireless communication system. Further, the cloud server causes the acquired information to be displayed on a display unit such as a display, whereby an administrator can easily recognize the communication quality of the wireless communication system.

Patent Literature 1 discloses that a capture device acquires and accumulates frames transmitted and received in wireless LAN communication between an access point and a terminal device for each of a plurality of predetermined measurement points. It is further disclosed that a measurement device analyzes characteristics of the wireless LAN communication, based on the frames accumulated in the capture device. Specifically, the measurement device calculates, for each terminal device, the number of data frames, an average bit rate, and the like at each measurement point. The terminal device performs wireless LAN communication with one access point, and switches the access point that performs wireless LAN communication depending on a radio wave condition or the like.

## CITATION LIST

### Patent Literature

[Patent Literature 1] Japanese Unexamined Patent Application Publication No. 2017-038332A

## SUMMARY OF INVENTION

### Technical Problem

The measurement device disclosed in Patent Literature 1 does not perform wireless LAN communication with a plurality of access points at the same time, but performs analysis of wireless LAN communication with respect to a terminal device that performs wireless LAN communication with one access point. In short, in the measurement device disclosed in Patent Literature 1, it is not assumed that analysis is performed on a terminal device that communicates with a plurality of access points by making a communication path redundant. Therefore, the measurement device disclosed in Patent Literature 1 has a problem that, when a terminal device having redundant communication paths is

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mixed in a wireless system, it is not possible to indicate an analysis result in consideration of the redundancy of communication paths.

An object of the present disclosure is to provide a server device, a monitoring method, and a program that are capable of performing analysis on a terminal device that simultaneously communicates with a plurality of access points by applying redundancy to a communication path.

## Solution to Problem

A server device according to a first aspect of the present disclosure includes: a communication unit configured to receive sensor information from a sensor device that collects packets transferred between a wireless terminal and a first access point and between the wireless terminal and a second access point, the sensor information being generated based on the collected packets; a management unit configured to manage the first access point and the second access point in association with the wireless terminal when the wireless terminal performs communication with the first access point and the second access point by applying redundancy to a communication path; and an analysis unit configured to generate information regarding a communication state of redundant communication paths, by using the sensor information regarding the first access point and the second access point.

A monitoring method according to a second aspect of the present disclosure includes: receiving sensor information from a sensor device that collects packets transferred between a wireless terminal and a first access point and between the wireless terminal and a second access point, the sensor information being generated based on the collected packets; managing the first access point and the second access point in association with the wireless terminal when the wireless terminal performs communication with the first access point and the second access point by applying redundancy to a communication path; and generating information regarding a communication state of redundant communication paths, by using the sensor information regarding the first access point and the second access point.

A program according to a third aspect of the present disclosure causes a computer to execute: receiving sensor information from a sensor device that collects packets transferred between a wireless terminal and a first access point and between the wireless terminal and a second access point, the sensor information being generated based on the collected packets; managing the first access point and the second access point in association with the wireless terminal when the wireless terminal performs communication with the first access point and the second access point by applying redundancy to a communication path; and generating information regarding a communication state of redundant communication paths, by using the sensor information regarding the first access point and the second access point.

## Advantageous Effects of Invention

According to the present disclosure, it is possible to provide a server device, a monitoring method, and a program that are capable of performing analysis on a terminal device that applies redundancy to a communication path and simultaneously communicates with a plurality of access points.

## BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a configuration diagram of a server device according to a first example embodiment;

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FIG. 2 is a configuration diagram of a visualization system according to a second example embodiment;

FIG. 3 is a configuration diagram of a wireless sensor according to the second example embodiment;

FIG. 4 is a configuration diagram of a cloud server 5 according to the second example embodiment;

FIG. 5 is a diagram illustrating a flow of transmission processing of sensor information in the wireless sensor according to the second example embodiment;

FIG. 6 is a diagram illustrating a flow of monitoring processing of a wireless LAN terminal in a cloud server 700 according to the second example embodiment; and

FIG. 7 is a configuration diagram of a sensor device, a wireless sensor, and a cloud server according to each example embodiment.

## EXAMPLE EMBODIMENT

### First Example Embodiment

Hereinafter, example embodiments of the present disclosure will be explained with reference to the drawings. A configuration example of a server device 10 according to a first example embodiment will be explained by using FIG. 1. The server device 10 may be a computer device that operates in such a manner that a processor executes a program stored in a memory.

The server device 10 includes a communication unit 11, a management unit 12, and an analysis unit 13. The communication unit 11, the management unit 12, and the analysis unit 13 may be software or modules that execute processing when a processor executes a program stored in a memory. Alternatively, the communication unit 11, the management unit 12, and the analysis unit 13 may be hardware such as a circuit or a chip.

The communication unit 11 receives sensor information from a sensor device. The sensor device collects packets transferred between the wireless terminal and a first access point and between the wireless terminal and a second access point. Further, the sensor device generates sensor information, based on the collected packets.

For example, the wireless terminal performs wireless Local Area Network (LAN) communication with the first access point and the second access point. The wireless terminal may be referred to as a slave unit in the wireless LAN communication, and the first access point and the second access point may be referred to as master units in the wireless LAN communication.

The packet may be referred to as a frame, data, or the like. The packet may be user data such as image data or moving image data, or may be control data. The user data may be referred to as a data frame, for example, and the control data may be referred to as a management frame or a control frame.

Collecting packets by the sensor device may be rephrased as capturing packets by the sensor device.

The sensor information may be, for example, Received Signal Strength Indicator (RSSI), a band occupancy rate, the number of packets transmitted by the access point in a predetermined period, and the number of packets transmitted by the wireless terminal in a predetermined period. Alternatively, the sensor information may be a throughput, a retransmission rate, or the like in communication between the wireless terminal and the access point.

The management unit 12 manages the first access point and the second access point in association with the wireless terminal when the wireless terminal performs communica-

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tion with the first access point and the second access point by applying redundancy to a communication path. The wireless terminal may apply redundancy to the communication path in such a manner that the wireless terminal has two wireless modules and performs data communication simultaneously with the first access point and the second access point. Alternatively, the wireless terminal may apply redundancy to a communication path in such a manner that the wireless terminal has two wireless modules and by switching the two wireless modules, data communication with the first access point and the second access point is performed. In these cases, for example, the wireless terminal communicates with the first access point by using one wireless module and with the second access point by using another wireless module.

The management unit 12 manages the first access point and the second access point in association with the wireless terminal as an access point in which the wireless terminal performs communication by applying redundancy to a communication path. The management unit 12 may update information of access points to be managed in association with the wireless terminal when the access point through which the wireless terminal performs communication by applying redundancy to a communication path is changed. For example, the management unit 12 may manage the first access point and the second access point in association with the wireless terminal by using a database. When the wireless terminal performs communication with the first access point and a third access point, the third access point is managed in association with the wireless terminal and the first access point, instead of the second access point.

The analysis unit 13 generates information regarding a communication state of the redundant communication path by using the sensor information regarding the first access point and the second access point. The information regarding the communication state may be information indicating the communication quality, or may be information indicating whether or not communication via the communication path is possible.

As explained above, when the wireless terminal performs communication with the first access point and the second access point by applying redundancy to the communication path, the server device 10 manages the first access point and the second access point in association with the wireless terminal. Accordingly, the server device 10 can manage the information regarding the communication state of the wireless terminal as analysis information regarding the redundant wireless terminal.

### Second Example Embodiment

Next, a configuration example of a visualization system according to a second example embodiment will be explained by using FIG. 2. The visualization system is a system in which a cloud server 700 monitors wireless LAN terminals 300 to 302 by using information acquired from wireless sensors 100 to 102.

The visualization system of FIG. 2 includes a base station 400, a core network 500, the Internet 600, and the cloud server 700. Further, the visualization system of FIG. 2 includes wireless sensors 100 to 102, APs 200 to 202, and wireless LAN terminals 300 to 302. The wireless sensors 100 to 102 are equivalent to the sensor device. The cloud server 700 is equivalent to the server device 10.

The base station 400 and the core network 500 are equivalent to a mobile network to be provided by a communication carrier.

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The wireless LAN terminals **300** to **302** are connected to any one of the APs **200** to **202**, which is a master unit, as slave units in wireless LAN communication, and perform wireless LAN communication. Further, it is illustrated that the wireless LAN terminal **300** and the wireless LAN terminal **301** have two wireless modules, and the wireless LAN terminal **302** has one wireless module. The wireless LAN terminal **301** and the wireless LAN terminal **302** have two wireless modules, and therefore, wireless LAN communication can be performed simultaneously with different APs. The wireless module may include, for example, an antenna, a modulator that modulates data, and a demodulator that demodulates data.

The wireless sensors **100** to **102** capture packets transmitted and received between the wireless LAN terminals **300** to **302** and the APs **200** to **202**.

The wireless sensors **100** to **102** perform bidirectional communication with the cloud server **700** via the base station **400**, the core network **500**, and the Internet **600**. The base station **400** may support, for example, Long Term Evolution (LTE), 5G, or local 5G as a radio communication standard. The base station **400** performs data communication by setting an LTE line, a 5G line, or a local 5G line to the wireless sensors **100** to **102** or the wireless terminal. Further, the wireless sensors **100** to **102** may perform communication with the Internet **600** via a wired line or Ethernet (registered trademark). Further, the cloud server **700** may be arranged in an intranet constructed in a specific company or the like.

Next, a configuration example of a wireless sensor **100** according to the second example embodiment will be explained by using FIG. 3. Further, since the wireless sensor **101** and the wireless sensor **102** have the same configuration as the wireless sensor **100**, detailed explanation thereof will be omitted.

The wireless sensor **100** includes a communication unit **111**, a collection unit **112**, and an extraction unit **113**. The communication unit **111**, the collection unit **112**, and the extraction unit **113** may be software or modules that execute processing when a processor executes a program stored in a memory. Alternatively, the communication unit **111**, the collection unit **112**, and the extraction unit **113** may be hardware such as a circuit or a chip.

The communication unit **111** performs communication with the base station **400**. The communication unit **111** performs communication with the base station **400** by using, for example, a wireless communication standard defined by 3GPP. Specifically, the communication unit **111** may perform communication with the base station **400** by using LTE. The communication unit **111** may be constituted of an antenna, a modulator, and a demodulator complying with a frequency of wireless communication with the base station **400**.

The communication unit **111** acquires a packet collection condition and extraction condition from the cloud server **700** via the base station **400**, the core network **500**, and the Internet **600**. The packet collection condition may be referred to as the packet capture condition. The communication unit **111** outputs the collection condition to the collection unit **112**, and outputs the extraction condition to the extraction unit **113**. The communication unit **111** may periodically receive the collection condition and the extraction condition from the cloud server **700**, or may irregularly receive a collection condition and an extraction condition that have been changed. The collection condition and the extraction condition are set or generated in the cloud server **700**.

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The collection unit **112** captures packets being transmitted and received between the wireless LAN terminals **300** to **302** and the APs **200** to **202** according to the collection conditions. The collection condition includes a frequency channel. The collection unit **112** captures a packet being transferred via the frequency channel included in the collection condition. In addition to the frequency channel, the collection condition may include a frequency band, a collection time, a collection period, the number of bytes, and the like.

The collection unit **112** may include an antenna, a modulator, and a demodulator complying with a frequency of wireless LAN communication with at least one of the APs **200** to **202**.

When two frequency channels are set as the collection condition, the collection unit **112** may capture packets of different frequency channels at the same time by using two antennas, modulators, and demodulators. Alternatively, when two frequency channels are set as the collection condition, the collection unit **112** may switch the frequency channels and capture packets of different frequency channels at predetermined time intervals by using one antenna, modulator, and demodulator.

The extraction unit **113** extracts a packet from among the packets captured by the collection unit **112** according to the extraction condition. Extracting a packet may be rephrased as selecting a packet. For example, the extraction condition may be a destination Basic Service Set Identifier (BSSID) or a source BSSID of the packet. The BSSID may be set with, for example, Media Access Control (MAC) addresses of APs. The extraction condition may be a destination IP address or a source IP address of the packet. The extraction condition may be a destination MAC address or a source MAC address of the packet. In short, the extraction unit **113** may extract a specific wireless LAN terminal, for example, a packet transmitted from the wireless LAN terminal **300**, from among the captured packets. Alternatively, the extraction unit **113** may extract all the packets transmitted to the AP **200**. The extraction unit **113** may designate addresses of a plurality of wireless LAN terminals as the extraction condition. In addition, an IP address or a MAC address related to a packet to be captured may be designated as the extraction condition, and an IP address or a MAC address related to a packet not to be captured may be designated, as the extraction condition.

The extraction unit **113** generates sensor information to be observed or measured from the extracted packet. A classification or a type and the like of the generated sensor information may be designated in the extraction condition. For example, the extraction unit **113** may generate address information indicating a source address or a destination address set in a header of the extracted packet. Alternatively, the extraction unit **113** may generate RSSI data observed from the extracted packet. The RSSI is not an RSSI related to a packet received by the wireless LAN terminal or the AP, but is an RSSI observed from a packet received or captured by the wireless sensor **100**.

Furthermore, the extraction unit **113** may generate throughput data and transmission rate data of the packets to be transmitted by the wireless LAN terminals **300** to **302**, by using the number of packets and a data length of the packets transmitted from the wireless LAN terminals **300** to **302** within a predetermined period. Furthermore, the extraction unit **113** may identify whether or not the packet is a retransmission packet by analyzing a header portion of the packet, and generate data indicating the number of retransmission packets. Further, the extraction unit **113** may gen-

erate data related to a band occupancy rate of the wireless LAN terminal **300** by using the maximum line speed between the AP and the wireless LAN terminal and the throughput data of the wireless LAN terminal **300**. Furthermore, the extraction unit **113** may generate data related to a retransmission rate of a packet by using the total number of received packets and the number of retransmission packets. The retransmission rate of the packet may be generated for each wireless LAN terminal or may be generated as data related to all wireless LAN terminals.

The communication unit **111** transmits information such as RSSI generated by the extraction unit **113** as sensor information to the cloud server **700** via the base station **400**, the core network **500**, and the Internet **600**.

Next, a configuration example of the cloud server **700** will be described by using FIG. 4. The cloud server **700** may be a computer device that operates when a processor executes a program stored in a memory. Further, the cloud server **700** may have a built-in memory for storing a database, or may be connected to a database server device via a network, a cable, or the like. The cloud server **700** saves data received from the wireless sensors **100** to **102** in a database.

The cloud server **700** includes a condition determination unit **701**, an analysis unit **702**, a communication unit **703**, and a management unit **704**. The condition determination unit **701**, the analysis unit **702**, the communication unit **703**, and the management unit **704** may be software or modules that execute processing when a processor executes a program stored in a memory. Alternatively, the condition determination unit **701**, the analysis unit **702**, the communication unit **703**, and the management unit **704** may be hardware such as a circuit or a chip.

The analysis unit **702** is equivalent to the analysis unit **13** of the server device **10**. The management unit **704** is equivalent to the management unit **12** of the server device **10**. The communication unit **703** is equivalent to the communication unit **11** of the server device **10**. Hereinafter, functions or operations of the cloud server **700** which are different from those of the server device **10** will be mainly explained, and detailed explanations of the same functions or operations as those of the server device **10** will be omitted.

The condition determination unit **701** displays information regarding the setting of the collection condition and the extraction condition on a display or the like that is used integrally with the cloud server **700**. An administrator of the cloud server **700** may confirm the information displayed on the display and input the collection condition and the extraction condition. For example, parameter information and the like that can be selected as the collection condition and the extraction condition may be displayed on the display.

The condition determination unit **701** outputs the collection condition and the extraction condition determined according to the input information to the management unit **704**. The condition determination unit **701** may save the collection condition and the extraction condition in the management unit **704** via the communication unit **703**. The communication unit **703** transmits the collection condition and the extraction condition being output from the condition determination unit **701** to at least one of the wireless sensors **100** to **102** via the Internet **600**, the core network **500**, and the base station **400**. The condition determination unit **701** may determine a different collection condition or extraction condition for each wireless sensor.

When the collection condition or the extraction condition is changed, the condition determination unit **701** may trans-

mit a change notification indicating that the collection condition or the extraction condition is changed to each wireless sensor. In this case, the wireless sensor that has received the change notification accesses the cloud server **700** and acquires a collection condition or extraction condition after change. Alternatively, the condition determination unit **701** may transmit the changed information to the wireless sensor without transmitting the change notification. Alternatively, the condition determination unit **701** may periodically transmit the collection condition or the extraction condition to each of the wireless sensors. For example, the condition determination unit **701** transmits the collection condition and the extraction condition to the wireless sensor via the communication unit **703**. Further, the condition determination unit **701** outputs the collection condition or extraction condition after change to the management unit **704** via the communication unit **703**. The management unit **704** stores the collection condition or extraction condition after change, which has been received.

The communication unit **703** receives sensor information from the wireless sensors **100** to **102**. The communication unit **703** saves the received sensor information in the management unit **704**. The analysis unit **702** processes the data saved in the management unit **704** into display data and displays the data on a display or the like.

When the wireless LAN terminal applies redundancy to the communication path and performs communication with a plurality of Aps, the management unit **704** manages the wireless LAN terminal and the AP of the connection destination in association with each other. For example, the wireless LAN terminal **300** can perform communication with the AP **200** and the AP **201** by using two wireless modules. The wireless LAN terminal **300** may perform communication with the AP **200** and the AP **201** at the same time by using two wireless modules, or may perform communication with the AP **200** and the AP **201** by switching the wireless modules to be used. A state in which the wireless LAN terminal **300** can perform communication with the AP may be rephrased as a state in which the wireless LAN terminal **300** is connected to the AP.

For example, the administrator or the like of the cloud server **700** groups the BSSID indicating the AP **200** and the BSSID indicating the AP **201**, which are communication destinations of the wireless LAN terminal **300**, and the wireless LAN terminal **300**, and inputs a name, ID, or the like for discriminating the group. The BSSID is information for discriminating respective wireless LAN communication areas formed by the AP **200** and the AP **201**. The respective MAC addresses of the AP **200** and the AP **201** may be set in the BSSID.

When the AP **200** and the AP **201** that are communication destinations of the wireless LAN terminal **300** are predetermined, the administrator or the like of the cloud server **700** may input the AP **200** and the AP **201** as APs to be grouped with the wireless LAN terminal **300**. Alternatively, the administrator or the like of the cloud server **700** may identify that the APs to which the wireless LAN terminal **300** is connected are the AP **200** and the AP **201** by checking the sensor information saved in the management unit **704**. In this case, the administrator or the like of the cloud server **700** may input the identified AP **200** and AP **201** as APs to be grouped with the wireless LAN terminal **300**. Alternatively, the management unit **704** may autonomously identify the APs to which the wireless LAN terminal **300** is connected, based on the saved or managed sensor information.

The analysis unit **702** generates quality information indicating the grouped wireless LAN terminals **300** from the

sensor information managed by the management unit **704** and received from the wireless sensors **100** to **102**. The quality information may be, for example, a RSSI related to the wireless LAN terminal **300**, a band occupancy rate, the number of packets transmitted by the wireless LAN terminal **300**, or the number of packets received by the wireless LAN terminal **300**. Alternatively, the quality information may be a throughput, a retransmission rate, or the like in communication between the wireless LAN terminal **300** and the AP **200** and between the wireless LAN terminal **300** and the AP **201**. Alternatively, the quality information may be Modulation Coding Scheme (MCS) related to the wireless LAN terminal **300**.

Next, a flow of processing of transmitting sensor information in the wireless sensor **100** according to the second example embodiment will be explained by using FIG. 5. First, the communication unit **111** receives setting information from the cloud server **700** (S101). The setting information includes at least one of a collection condition and an extraction condition. For example, the communication unit **111** may receive a newly set collection condition and extraction condition, or may receive a changed collection condition or extraction condition.

Next, the collection unit **112** captures a packet according to the collection condition (S102). For example, the collection unit **112** captures a packet transferred via a frequency channel included in the collection condition. Next, the extraction unit **113** generates sensor information observed from the extracted packet according to the extraction condition (S103). For example, the extraction unit **113** generates RSSI data as sensor information.

Next, the communication unit **111** transmits the sensor information generated by the extraction unit **113** to the cloud server **700** (S104).

Next, the communication unit **111** determines whether or not a change notification of the setting information has been received from the cloud server **700** (S105). When the communication unit **111** receives the change notification of the setting information, the processing of step S101 and subsequent steps are repeated. When the communication unit **111** has not received the change notification of the setting information, the processing of step S102 and subsequent steps are repeated.

Next, a flow of monitoring processing of the wireless LAN terminal in the cloud server **700** according to the second example embodiment will be explained by using FIG. 6. First, the management unit **704** performs group settings related to the wireless LAN terminal **300** and the like (S201). For example, when there are two BSSID that are communication destinations or connection destinations of the wireless LAN terminal **300**, the management unit **704** manages the two BSSIDs and the wireless LAN terminal **300** as one group. For example, the management unit **704** manages a group by using a name, an ID, or the like that discriminates the group. An administrator or the like of the cloud server **700** may input the name, ID, or the like of the group via an input device of the cloud server **700**. In addition, the management unit **704** may add a BSSID, a frequency channel to be a new monitoring target, or the like to a group that has already been set.

Next, the analysis unit **702** refers to the sensor information saved in the management unit **704**, and extracts the sensor information regarding the wireless LAN terminal that performs communication with a plurality of access points by applying redundancy to the communication path (S202). For example, the analysis unit **702** extracts sensor information regarding the wireless LAN terminal **300**. The analysis unit

**702** may extract sensor information regarding the wireless LAN terminal designated by the administrator of the cloud server **700** or the like. Alternatively, the analysis unit **702** may identify a wireless LAN terminal that performs communication with a plurality of access points by applying redundancy to the communication path, and extract sensor information regarding the identified wireless LAN terminal. For example, the analysis unit **702** may identify, from the sensor information, a wireless LAN terminal that performs communication with a plurality of access points in a predetermined period as a wireless LAN terminal that performs communication with a plurality of access points by applying redundancy to the communication path.

Next, the analysis unit **702** generates display data related to the group (S203). The display data are data displayed on a display or the like that is used integrally with the cloud server **700**. For example, the analysis unit **702** generates quality information regarding the wireless LAN terminal **300** as display data. The quality information may be, for example, a RSSI related to the wireless LAN terminal **300**, a band occupancy rate, the number of packets transmitted by the wireless LAN terminal **300**, or the number of packets received by the wireless LAN terminal **300**. Alternatively, the quality information may be a throughput, a retransmission rate, or the like in communication between the wireless LAN terminal **300** and the AP **200** and between the wireless LAN terminal **300** and the AP **201**. Alternatively, the quality information may be an MCS related to the wireless LAN terminal **300**.

In addition, when the wireless LAN terminal **300** simultaneously performs data communication by using a redundant communication path, the analysis unit **702** may generate a combined value, an average value, a maximum value, a minimum value, and the like of the quality information as display data. The simultaneous data communication is to perform data communication with the AP **200** and the AP **201** at substantially the same timing by using two wireless modules included in the wireless LAN terminal **300**. Substantially the same timing does not only include exactly the same timing, but also includes a case where the wireless LAN terminal **300** performs data communication by using two wireless modules within a predetermined period.

The analysis unit **702** may generate, as display data, a combined value, an average value, a maximum value, a minimum value, and the like of a throughput of data communication between the wireless LAN terminal **300** and the AP **200** and a throughput of data communication between the wireless LAN terminal **300** and the AP **201**.

Alternatively, in a case where the communication path to be used for data communication is switched among the redundant communication paths and used, the analysis unit **702** may generate, in addition to the quality information, the number of times of switching the communication path, a switching timing, an establishment status of a communication path for which data communication is not performed, and the like. The establishment status of the communication path for which data communication is not performed may be, for example, information indicating whether or not data communication is performed in the communication path for which data communication is not performed. Specifically, it may be information regarding whether or not a wireless link is established between the wireless LAN terminal **300** and an AP on the communication path for which data communication is not performed.

Next, the analysis unit **702** displays the display data on a display that is used integrally with the cloud server **700**, and saves the display data in the management unit **704** (S204).

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Thereafter, the analysis unit **702** may repeat the processing in step **S202** and subsequent steps, and display the quality information regarding the wireless LAN terminal **300** on the display as time-series data.

Further, the analysis unit **702** displays the display data saved in the management unit **704** on the display in accordance with an operation of a page displayed on the display, which is performed by the administrator of the cloud server **700** or the like. For example, when the administrator or the like inputs information indicating that the wireless LAN terminal **300** is to be displayed, the analysis unit **702** may simultaneously display quality information regarding data communication between the wireless LAN terminal **300** and the AP **200** and between the wireless LAN terminal **300** and the AP **201**. Alternatively, the analysis unit **702** may display a combined value or the like of quality information regarding data communication between the wireless LAN terminal **300** and the AP **200** and between the wireless LAN terminal **300** and the AP **201**. Alternatively, the analysis unit **702** may display only the quality information regarding one of the APs in which the wireless LAN terminal **300** performs data communication, and may display the quality information regarding another AP in such a way that the quality information regarding the another AP can be selected.

As described above, the cloud server **700** according to the second example embodiment can manage a wireless LAN terminal having a redundant communication path and a plurality of APs that are communication destinations of the wireless LAN terminal in association with each other. Further, the cloud server **700** can calculate a combined value, an average value, and the like of the quality information regarding the wireless LAN terminal having a redundant communication path. Thus, the cloud server **700** can analyze the quality information of the wireless LAN terminal in consideration of the redundancy of the communication path.

FIG. 7 is a block diagram illustrating a configuration example of the server device **10**, the wireless sensor **100**, and the cloud server **700** (hereinafter, referred to as the server device **10** and the like). Referring to FIG. 7, the server device **10** and the like include a network interface **1201**, a processor **1202**, and a memory **1203**. The network interface **1201** is used for communicating with a network node (e.g., eNB, MME, P-GW). The network interface **1201** may include, for example, a network interface card (NIC) compliant with IEEE 802.3 series. Herein, eNB represents evolved Node B, MME represents Mobility Management Entity, and P-GW represents Packet Data Network Gateway. IEEE represents Institute of Electrical and Electronics Engineers.

The processor **1202** reads and executes software (a computer program) from the memory **1203** and performs the processing of the server device **10** and the like that is explained by using the flowchart in the above-described example embodiment. The processor **1202** may be, for example, a microprocessor, an MPU, or a CPU. The processor **1202** may include a plurality of processors.

The memory **1203** is constituted of a combination of a volatile memory and a non-volatile memory. The memory **1203** may include a storage located remotely from the processor **1202**. In this case, the processor **1202** may access the memory **1203** via Input/Output (I/O) interfaces that are not illustrated.

In the example of FIG. 7, the memory **1203** is used for storing a group of software modules. The processor **1202** reads a group of these software modules from the memory **1203** and execute the software modules, thereby enabling to

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perform the processing of the server device **10** and the like explained in the above-described example embodiment.

As explained by using FIG. 7, each of the processors included in the server device **10** and the like in the above-described example embodiment executes one program or a plurality of programs including a group of instructions for causing a computer to execute the algorithm explained by using the drawings.

In the examples described above, the program may be stored and provided to the computer by using various types of non-transitory computer-readable media. Non-transitory computer-readable media include various types of tangible storage media. Examples of the non-transitory computer-readable media include magnetic recording media (e.g., flexible disks, magnetic tapes, hard disk drives), magneto-optical recording media (e.g., magneto-optical disks), Read Only Memory (CD-ROM), CD-R, CD-R/W, and semiconductor memory (e.g., mask ROM, Programmable ROM (PROM), Erasable PROM (EPROM), flash ROM, Random Access Memory (RAM)). The program may also be provided to the computer by various types of transitory computer readable media. Examples of the transitory computer readable media include electrical signals, optical signals, and electromagnetic waves. The transitory computer readable medium can provide the program to the computer via a wired communication path such as an electric wire and an optical fiber, or a wireless communication path.

The present disclosure is not limited to the above-described example embodiments, and can be appropriately modified without departing from the spirit of the present disclosure.

Although the present invention has been explained with reference to the example embodiments, the present invention is not limited to the above. Various modifications that can be understood by a person skilled in the art within the scope of the invention can be made to the configuration and details of the present invention.

This application is based upon and claims the benefit of priority from Japanese patent application No. 2020-114079, filed on Jul. 1, 2020, the disclosure of which is incorporated herein in its entirety.

## REFERENCE SIGNS LIST

**10** SERVER DEVICE  
**11** COMMUNICATION UNIT  
**12** MANAGEMENT UNIT  
**13** ANALYSIS UNIT  
**100** WIRELESS SENSOR  
**101** WIRELESS SENSOR  
**102** WIRELESS SENSOR  
**111** COMMUNICATION UNIT  
**112** COLLECTION UNIT  
**113** EXTRACTION UNIT  
**200** AP  
**201** AP  
**202** AP  
**300** WIRELESS LAN TERMINAL  
**301** WIRELESS LAN TERMINAL  
**302** WIRELESS LAN TERMINAL  
**400** BASE STATION  
**500** CORE NETWORK  
**600** INTERNET  
**700** CLOUD SERVER  
**701** CONDITION DETERMINATION UNIT  
**702** ANALYSIS UNIT  
**703** COMMUNICATION UNIT  
**704** MANAGEMENT UNIT

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What is claimed is:

1. A server device comprising:  
at least one memory storing instructions, and  
at least one processor configured to execute the instructions to;  
receive sensor information from a sensor device that  
collects packets transferred between a wireless terminal  
and a first access point and between the wireless  
terminal and a second access point, the sensor information  
being generated based on the collected packets;  
manage the first access point and the second access point  
in association with the wireless terminal when the  
wireless terminal performs communication with the  
first access point and the second access point by  
applying redundancy to a communication path; and  
generate information regarding a communication state of  
redundant communication paths by using the sensor  
information regarding the first access point and the  
second access point.
2. The server device according to claim 1, wherein the at  
least one processor is further configured to execute the  
instructions to manage a first BSSID used by the wireless  
terminal for communication with a first access point and a  
second BSSID used by the wireless terminal for communication  
with a second access point in association with each  
other.
3. The server device according to claim 1, wherein the at  
least one processor is further configured to execute the  
instructions to combine first sensor information regarding a  
first communication path with the first access point and  
second sensor information regarding a second communication  
path with the second access point, average the first  
sensor information and the second sensor information, or  
select a maximum value or a minimum value of the first  
sensor information and the second sensor information, and  
generate information regarding the communication state.
4. The server device according to claim 3, wherein, when  
the wireless terminal performs communication by switching  
between the first communication path and the second communication  
path, the at least one processor is further configured to  
execute the instructions to generate information indicating  
the number of times of switching in a predetermined period  
as information regarding the communication state.
5. The server device according to claim 3, wherein, when  
the wireless terminal performs communication by switching  
between the first communication path and the second communication  
path, the at least one processor is further configured to  
execute the instructions to generate information indicating  
an establishment state of a wireless link in the first  
communication path or the second communication path that  
is not used for communication as information regarding the  
communication state.

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6. A monitoring method in a server device, comprising:  
receiving sensor information from a sensor device that  
collects packets transferred between a wireless terminal  
and a first access point and between the wireless  
terminal and a second access point, the sensor information  
being generated based on the collected packets;  
managing the first access point and the second access  
point in association with the wireless terminal when the  
wireless terminal performs communication with the  
first access point and the second access point by  
applying redundancy to a communication path; and  
generating information regarding a communication state  
of redundant communication paths by using the sensor  
information regarding the first access point and the  
second access point.
7. The monitoring method according to claim 6, further  
comprising, when managing the first access point and the  
second access point in association with the wireless terminal,  
managing a first BSSID used by the wireless terminal for  
communication with a first access point and a second  
BSSID used by the wireless terminal for communication  
with a second access point in association with each other.
8. The monitoring method according to claim 6, further  
comprising, when generating information regarding the  
communication state, combining first sensor information  
regarding a first communication path with the first access  
point and second sensor information regarding a second  
communication path with the second access point, averaging  
the first sensor information and the second sensor information  
or selecting a maximum value or a minimum value of the  
first sensor information and the second sensor information,  
and generating information regarding the communication  
state.
9. A non-transitory computer-readable medium storing a  
program that causes a computer to execute:  
receiving sensor information from a sensor device that  
collects packets transferred between a wireless terminal  
and a first access point and between the wireless  
terminal and a second access point, the sensor information  
being generated based on the collected packets;  
managing the first access point and the second access  
point in association with the wireless terminal when the  
wireless terminal performs communication with the  
first access point and the second access point by  
applying redundancy to a communication path; and  
generating information regarding a communication state  
of redundant communication paths by using the sensor  
information regarding the first access point and the  
second access point.

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